A16728W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4. NUCLEAR AND PARTICLE PHYSICS

TRINITY TERM 2023

Monday, 05 June, 2.30 pm – 4.30 pm

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

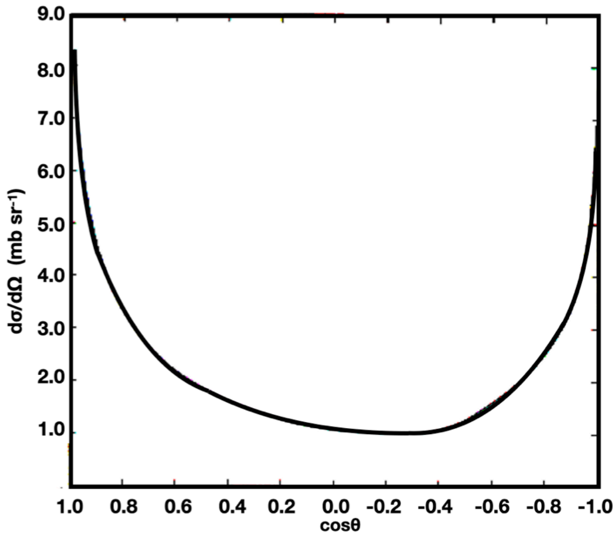
The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Explain the origin of isospin and whether it is an approximate or exact symmetry. What are the reasons behind its conservation or violation in the strong and weak interaction? Is isospin symmetry preserved for electromagnetic interactions? [4]

(b) Derive the possible values of total isospin for the systems (i) $p + \pi^+$ and (ii) $p + \pi^-$. [2]

(c) Consider the following plot of the differential cross section for $\sim$1 GeV protons elastically scattering off stationary neutrons:



where θ is the scattering angle in the centre of mass frame. What is the reason for the nearly symmetric forward and backward peaks? Draw quark flow diagrams for reactions representing each of these peaks, demonstrating how they can be couched in terms of an effective pion exchange. Explicitly show in the quark flow diagrams how colour is conserved. [6]

(d) Show that when the internal momentum, q, of nucleons is taken into account, the threshold energy, E'_{min}, for producing antiprotons in the collision of protons with nuclei is given approximately by:

$$E'_{\mathrm{min}} \simeq E_{\mathrm{min}}(1 - q/m_p c)$$

where E_{min} is the threshold energy for collisions with free protons and m_p is the proton mass. Estimate the value of q. [4]

(e) Show that the process $p + \overline{p} \to \pi^+ + \pi^-$ can be represented in a quark flow diagram by the effective exchange of either a doubly charged or neutral hadron. Identify the possible hadrons. [4]

(f) What difference in the mediators for strong vs electromagnetic interactions is responsible for the phenomena of confinement and asymptotic freedom? [2]

(g) Is the following process allowed? If so, draw a representative dominant Feynman diagram. If not, explain what principle is violated.

$$\pi^- + p \to \pi^- + \pi^0 + \pi^+ + n$$ [3]

2. (a) Draw leading order Feynman diagrams for the production of W bosons by three different exchange modes in $e^+ \, e^-$ collisions. [3]

(b) List the different decay modes of the W boson. For a collider detector surrounding the interaction region of an $e^+ \, e^-$ collider, describe the characteristic detection signatures of a single W boson decay for these different modes. [5]

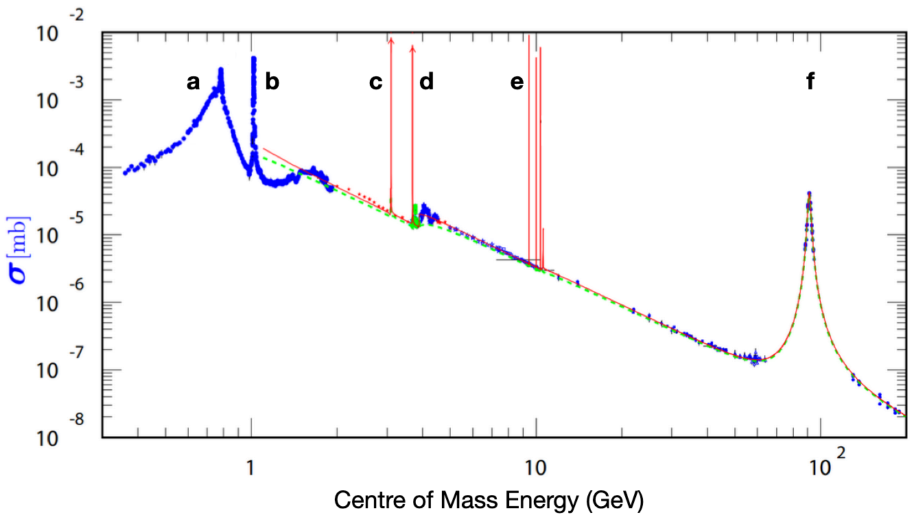
(c) For particles colliding to produce a new intermediary particle that subsequently decays, the relativistic form of the Breit-Wigner cross section is given by:

$$\sigma = \frac{\alpha \Gamma_i \Gamma_f}{(E^2 - M^2)^2 + M^2 \Gamma^2}.$$

(i) Define each term in this formula. What variables does α depend upon? [2]

(ii) Show that, for energies very close to the resonance, the familiar non-relativistic expression can be recovered. [3]

(d) The plot below shows the cross section for $e^+ e^- \rightarrow$ hadrons:



(i) Identify the nature of the main resonances labelled a-f. [6]

(ii) Explain the overall slope and the reason for the discontinuity near d. [2]

(iii) Explain the reason for the breadth of peaks at a and f. [4]

3. (a) Draw leading order quark-level Feynman diagrams for the processes of beta decay and 'inverse' beta decay (*i.e.* $\bar{\nu}_e + p \rightarrow n + e^+$), showing how the processes are related. [2]

(b) Prior to the development of electroweak theory, Bruno Pontecorvo used a simple dimensional argument to estimate the weak interaction cross section based on the characteristic weak interaction timescale for the decay of a free neutron, which is $\sim$880 seconds.

(i) By considering the wavepacket for a 1 MeV anti-neutrino and the time it spends in the presence of a proton, estimate the nominal cross section for inverse beta decay at this energy. [3]

(ii) Find the energy dependence of the inverse beta decay cross section for interactions on a fixed target nucleus. [3]

(iii) Derive the threshold energy for such inverse beta decay interactions. [3]

(c) List the main assumptions of the liquid drop model and briefly explain the terms in the semi-empirical mass formula (SEMF) for binding energy (in MeV):

$$\mathrm{BE} = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta \,,$$

where a_V=15.75, $a_S = 17.8$, $a_C = 0.711$, $a_A = 23.7$, $\delta = (11.18/\sqrt{A},\ 0,\ -11.18/\sqrt{A})$. [4]

(d) Some nuclei are stable against beta decay but unstable to double beta decay, in which two neutrons simultaneously decay as a single quantum event. What aspect of the nuclear mass model is responsible for this behaviour? Estimate the Q value for the double beta decay of $^{130}_{52}$Te (note: the value given by the SEMF is about 6.5 MeV too low). [3]

(e) What aspects of the Standard Model would prevent massless neutrinos from being converted into anti-neutrinos? If such a conversion were possible, with massive neutrinos, then a process of neutrinoless double beta decay (*i.e.* with only electrons emitted) could occur via neutrino exchange. Thus, the electrons would carry the full endpoint energy. The half-life for neutrinoless double beta decay would be given by the following expression:

$$\left(T^{0\nu}_{1/2}\right)^{-1} = G^{0\nu}|M^{0\nu}|^2 m^2_{\beta\beta}$$

where $G^{0\nu}$ is a phase space factor for two electrons to escape from a given nucleus, $M^{0\nu}$ is the nuclear matrix element for the transition and $m_{\beta\beta}$ is the effective neutrino mass for the process in units of the electron rest mass. Justify the form of this expression. [2]

(f) For the double beta decay isotope ^{130}Te, $G^{0\nu} = 5.5 \times 10^{-15}$yrs^{-1} and a typical estimated value for $M^{0\nu}$ is 4. Assume that 1 tonne of the isotope is viewed by a detector for 1 year, and no events are observed at the full endpoint energy. Assuming the observation obeys Poisson counting statistics, place an upper bound on the value of $m_{\beta\beta}$ with 90% certainty. [5]

4. (a) Consider a beam of N particles fired during a time $\mathrm{d}t$ at the face of a thin block of area A and thickness $\mathrm{d}x$ (hence a volume $V = A\mathrm{d}x$). The block contains N_T equivalent hard scattering targets, each of projected cross sectional area σ.

(i) Derive the relationship

$$\sigma = \frac{R_S V}{N N_T v},$$

making clear your definition of the scattering rate R_S and stating what the variable v here refers to. Identify which part of your formulae is the rate which could in principle be obtained by the Fermi Golden Rule. [3]

(ii) For a thick block of length x, the number of scattered particles can be written as $N_{\mathrm{scatt}} = N[1 - \exp(\alpha)]$. Find the expression for α. [3]

(b) Assume we can approximate nucleon scattering by a spherical square well potential of the form: $V(r) = -V_0$ for $r < a$, and $V(r) = 0$ for $r \geq a$.

(i) Use the first order Born approximation to derive a differential cross section of the form:

$$A[\sin(qa) - B\cos(qa)]^2$$

where q is the momentum transfer. Find the factors A and B. [4]

(ii) In the low energy limit $ka \ll 1$, where k is the wave number of the incoming particle, show that the total cross section can be expressed in the form $4\pi R^2$. Determine the factor R. [4]

(c) Consider the nuclei of a graphite moderator to be pure ^{12}C, with a number density of 9×10^{28} nuclei/m^3. For neutron energies less than 2 MeV, the scattering is elastic, with an approximately constant cross section of approximately 4.5 barns.

(i) A neutron with non-relativistic kinetic energy E_0 interacts with the moderator. In the centre of mass system, the scattering is isotropic. Show that the average neutron kinetic energy in the lab frame after each collision is:

$$E = \frac{M^2 + m_n^2}{(M + m_n)^2} E_0 \equiv \alpha E_0$$

where M is the mass of the ^{12}C nucleus. [4]

(ii) Estimate the mean number of collisions required to reduce the energy of a 2 MeV fission neutron to a thermal energy of 0.1 eV. [1]

(iii) Derive an expression for the mean time between collisions for a neutron of energy E. [2]

(iv) Give an expression for the average rate of energy loss. Hence, find the time it takes to thermalise a 2 MeV neutron (*i.e.* to reduce its energy to 0.1 eV). [4]